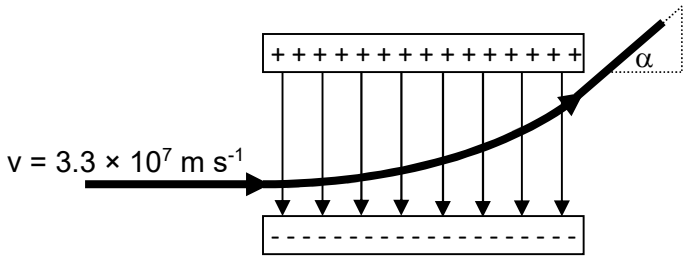
Review paper

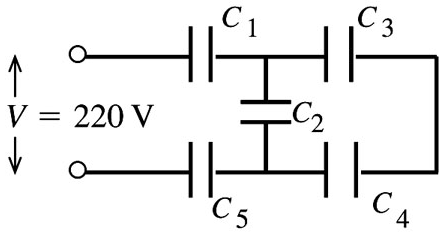
1. An electron gun creates a beam of electrons moving horizontally with a speed of 3.3×10^7 m s^{-1}. The electrons enter two parallel electrodes of length 2.0 cm, where the electric field is $E = 5.0 \times 10^4$ N C^{-1} downwards. In which direction and by what angle is the electron beam deflected by the electrodes?



2. A positive charge q is fixed at the point x = 0, y = 0. A negative charge -2q is fixed at the point x = a and y = 0. (a) Show the positions of the charges in a diagram. (b) Derive an expression for the potential V at points on the x-axis as a function of the coordinate x (take V to be zero at an infinite distance from the charges).(c) At which positions on the x-axis is V = 0?

3. Three point charges are arranged along the *x*-axis. Charge $q_1 = +3.00$ μC is at the origin and charge $q_2 = -5.00$ μC is at x = 0.200 m. There is a third charge $q_3 = -8.00$ μC located on the *x*-axis. Where is q_3 located so that the net force on q_1 is 7.00 N in the negative *x*-direction?

4. In the below figure, $C_1 = C_5 = 8.4$ μF and $C_2 = C_3 = C_4 = 4.2$ μF. The applied potential is $V_{ab} = 220$ V. (a) What is the equivalent capacitance of the network between points a and b? (b) Calculate the charge on each capacitor and the potential difference across each capacitor.



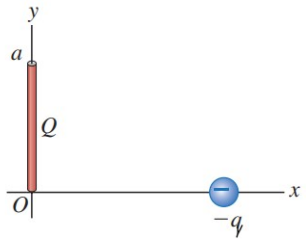
5. A capacitor has parallel plates of area 12 cm^2 and separated by 2.0 mm. The space between the plates is filled with polystyrene (K=2.6). (a) Find the permittivity of polystyrene. (b) Find the maximum permissible voltage across the capacitor to avoid dielectric breakdown (The dielectric strength is E= 2 x 10^7 V/m). (c) When the voltage equals the value found in part (b), find the surface charge density on each plate and the induced surface charge density on the surface of the dielectric.

6. A hollow, conducting sphere with an outer radius of 0.250 m and an inner radius of 0.200 m has a uniform surface charge density of +6.37 x 10^{-6} C/m^2. A charge of -0.500 μC is now introduced into the cavity inside the sphere. (a) What is the new charge density on the outside of the sphere? (b) Calculate the strength of the electric field just outside the sphere. (c) What is the electric flux through a spherical surface just inside the inner surface of the sphere?

7. In a certain region of space the electric potential is given by where V= Ax^2y − Bxy^2 where A = 5.00 V/m^3 and B = 8.00 V/m^3. Calculate the magnitude and direction of the electric field at the point in the region that has coordinates x = 2.00 m, y = 0.400 m, and z = 0.

8. (a) What must the charge (sign and magnitude) of a 1.45 g particle be for it to remain stationary when placed in a downward-directed electric field of magnitude 650 N/C?(b) What is the magnitude of an electric field in which the electric force on a proton is equal in magnitude to its weight?

9. In the figure, positive charge Q is distributed uniformly along the positive y-axis between y= 0 and $y = a$. A negative point charge $-q$ lies on the positive x-axis, a distance x from the origin. (a) Calculate the electric field produced by the charge distribution at points on the positive x-axis. (b) Calculate the force that the charge distribution exerts on $-q$. (c) When $x >> a$, show that the charge distribution acts like a point charge.



10. Three charges are at the corners of an isosceles triangle as shown in the figure. The ±5.00 μC charges form a dipole. (a) Find the force (magnitude and direction) the −10.00 μC charge exerts on the dipole. (b) For an axis perpendicular to the line connecting the ±5.00 μC charges at the midpoint of this line, find the torque (magnitude and direction) exerted on the dipole by the −10.00 μC charge.

